Hybrid Laser-Induced Graphene/Zinc Conductors for Sustainable Electronic Devices

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1-Page Extended ABSTRACT

Transient and sustainable electronics are becoming increasingly relevant for applications in disposable sensing, smart packaging, soft wireless systems, and bioresorbable devices. Zinc is of particular interest as a conductive material because it combines electrical functionality with degradability and compatibility with transient electronic systems. Recent studies have demonstrated Zn-based printed conductors, resistive and capacitive sensors, and radio-frequency components compatible with transient substrates. However, key challenges remain in achieving high electrical conductivity and long-term stability. Most Zn-based approaches rely on particle deposition followed by chemical, photonic, or hybrid sintering, with conductor performance strongly depending on particle coalescence, oxide removal, and post-processing conditions [1-3].

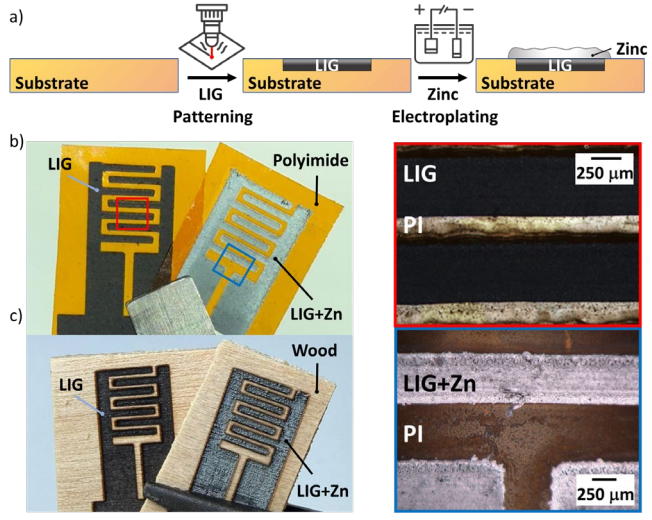


Fig. 1 (a) Schematic illustration of the fabrication process: laser-induced graphene (LIG) is first patterned on the substrate and subsequently used as a seed layer for zinc electroplating. (b-c) Optical images of an interdigitated capacitor before and after Zn electroplating on PI (b) and wood (c) substrates.

This work describes a route to pattern Zn conductors based on laser-induced graphene (LIG) used as a conductive seed layer for zinc electroplating. In this approach, conductive graphitic patterns are first generated on polyimide (PI) and birch wood by direct laser writing using a Trotec Speedy 300 CO_2 laser system (maximum power: 150 W), and then used as electrically active template for subsequent Zn electrodeposition (Figure 1a). Electroplating was performed in an aqueous 0.5 M zinc sulfate heptahydrate bath, using a power generator to provide a constant current during plating.

The process combines maskless and digitally controlled laser patterning with the formation of a metallic Zn coating, producing conductive features with defined geometry and tunable thickness.

The proposed architecture offers several advantages over direct Zn particles printing and sintering. First, the conductive pathway is established during the laser-writing, enabling direct and localized pattern definition without the need to pattern a separate seed layer. Second, the electroplating step allows Zn deposition to be adjusted through plating conditions, providing control over conductor thickness and electrical performances.

The LIG layer was optimized on PI following [4], using 17% power, 6% speed, and a −2 mm offset, to reach to a sheet resistance of about 18 Ω/□ while on wood a minimum sheet resistance of about 15 Ω/□ was obtained using 16% power, 3% speed, and a 3 mm offset; both sets of conditions were suitable for zinc electroplating. The design, dimensions, and geometry of the patterned structures also influence the plating process and therefore must be carefully adjusted. By tuning the plating current and deposition time, Zn thicknesses ranging from a few to several tens of micrometers can be obtained. For the electrodes presented in Figure 1, the electroplated Zn layer reached a thickness of a few tens of micrometers with a lower sheet resistance obtained, R_S, of 6.5 × 10^{-2} Ω/□ on PI and R_S, of 2.2 × 10^{-1} Ω/□ on wood. The mechanical adhesion of the films was confirmed by a scotch test and bending experiments are in progress to evaluate their flexibility.

The process was successfully demonstrated on both PI and wood, showing its applicability to polymeric and naturally derived substrates. LIG-assisted Zn electroplating yielded patterned conductors with low sheet resistance. This combination of digital patterning and controllable metallization is attractive for sustainable and transient electronics.

References

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